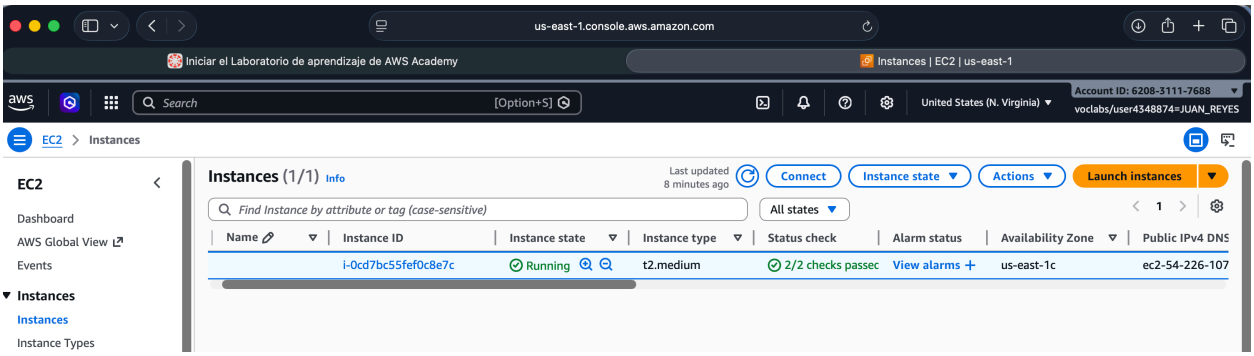
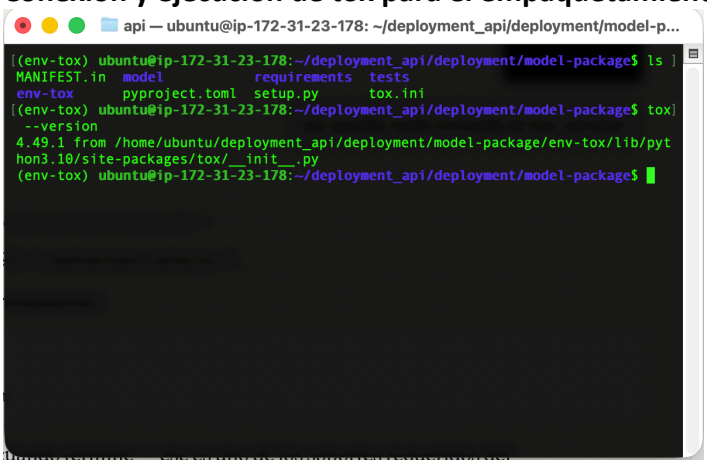
Microproyecto: entregable 3

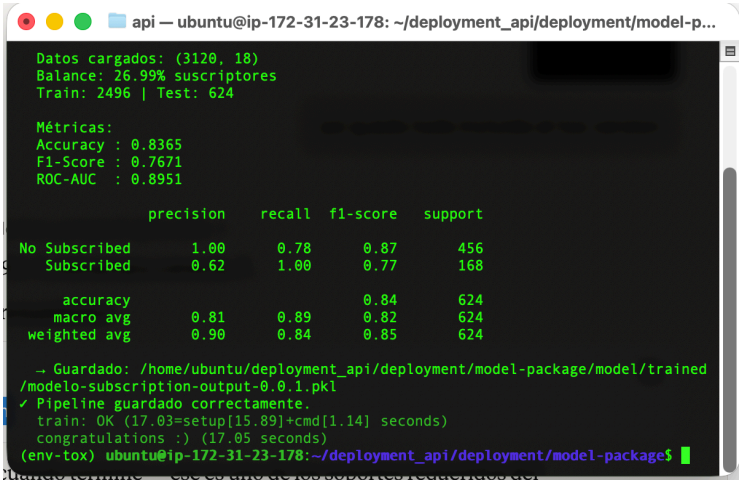
Evidencia de la instancia de ec2



Conexion y ejecucion de tox para el empaquetamiento del modelo:



Comando tox run -e train



Pruebas unitarias:

```
api — ubuntu@ip-172-31-23-178: ~/deployment_api/deployment/model-p...
/modelo-subscription-output-0.0.1.pkl
✓ Pipeline guardado correctamente.
test_package: commands[1]> pytest -s -vv tests/
============================= test session starts ==============================
platform linux -- Python 3.10.12, pytest-9.0.2, pluggy-1.6.0 -- /home/ubuntu/dep
loyment_api/deployment/model-package/.tox/test_package/bin/python
cachedir: .tox/test_package/.pytest_cache
rootdir: /home/ubuntu/deployment_api/deployment/model-package
configfile: pyproject.toml
collected 5 items

tests/test_prediction.py::test_prediction_count PASSED
tests/test_prediction.py::test_predictions_are_binary PASSED
tests/test_prediction.py::test_probabilities_in_range PASSED
tests/test_prediction.py::test_model_version PASSED
tests/test_prediction.py::test_single_record
  Predicción:   1
  Probabilidad: 0.7519
PASSED

============================== 5 passed in 0.22s ===============================
  test_package: OK (19.42=setup[16.79]+cmd[1.12,1.51] seconds)
  congratulations :) (19.44 seconds)
(env-tox) ubuntu@ip-172-31-23-178:~/deployment_api/deployment/model-package$
```

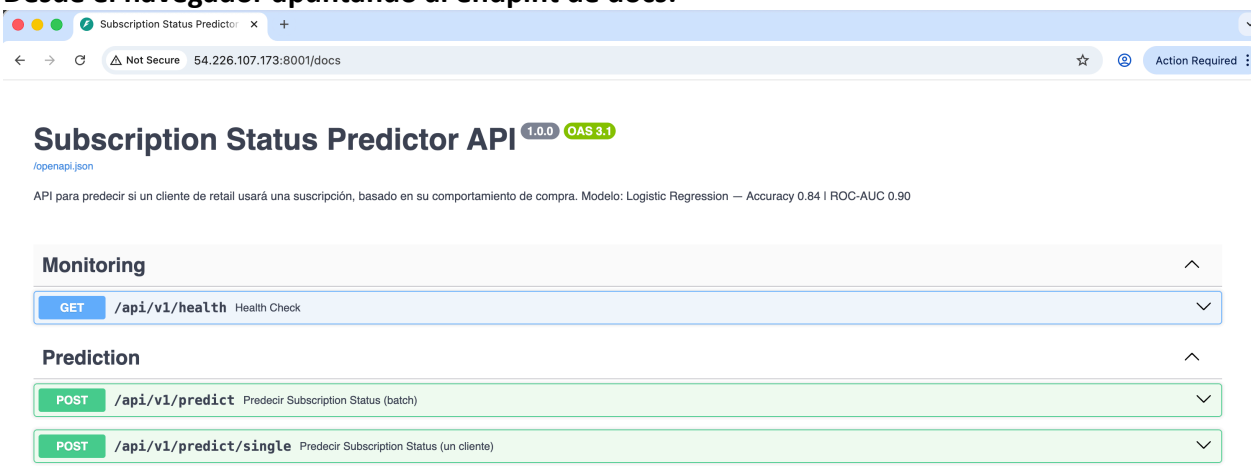
Construccion del paquete con build:

```
api — ubuntu@ip-172-31-23-178: ~/deployment_api/deployment/model-package — ssh -i llave/deploym...
adding 'model/VERSION'
adding 'model/__init__.py'
adding 'model/config.yml'
adding 'model/pipeline.py'
adding 'model/predict.py'
adding 'model/train_pipeline.py'
adding 'model/config/__init__.py'
adding 'model/config/core.py'
adding 'model/datasets/__init__.py'
adding 'model/datasets/shopping_test.csv'
adding 'model/datasets/shopping_train.csv'
adding 'model/processing/__init__.py'
adding 'model/processing/data_manager.py'
adding 'model/processing/features.py'
adding 'model/processing/validation.py'
adding 'model/trained/__init__.py'
adding 'model/trained/modelo-subscription-output-0.0.1.pkl'
adding 'model_subscription-0.0.1.dist-info/METADATA'
adding 'model_subscription-0.0.1.dist-info/WHEEL'
adding 'model_subscription-0.0.1.dist-info/top_level.txt'
adding 'model_subscription-0.0.1.dist-info/RECORD'
removing build/bdist.linux-x86_64/wheel
Successfully built model_subscription-0.0.1.tar.gz and model_subscription-0.0.1-py3-none-any.whl
(env-tox) ubuntu@ip-172-31-23-178:~/deployment_api/deployment/model-package$
```

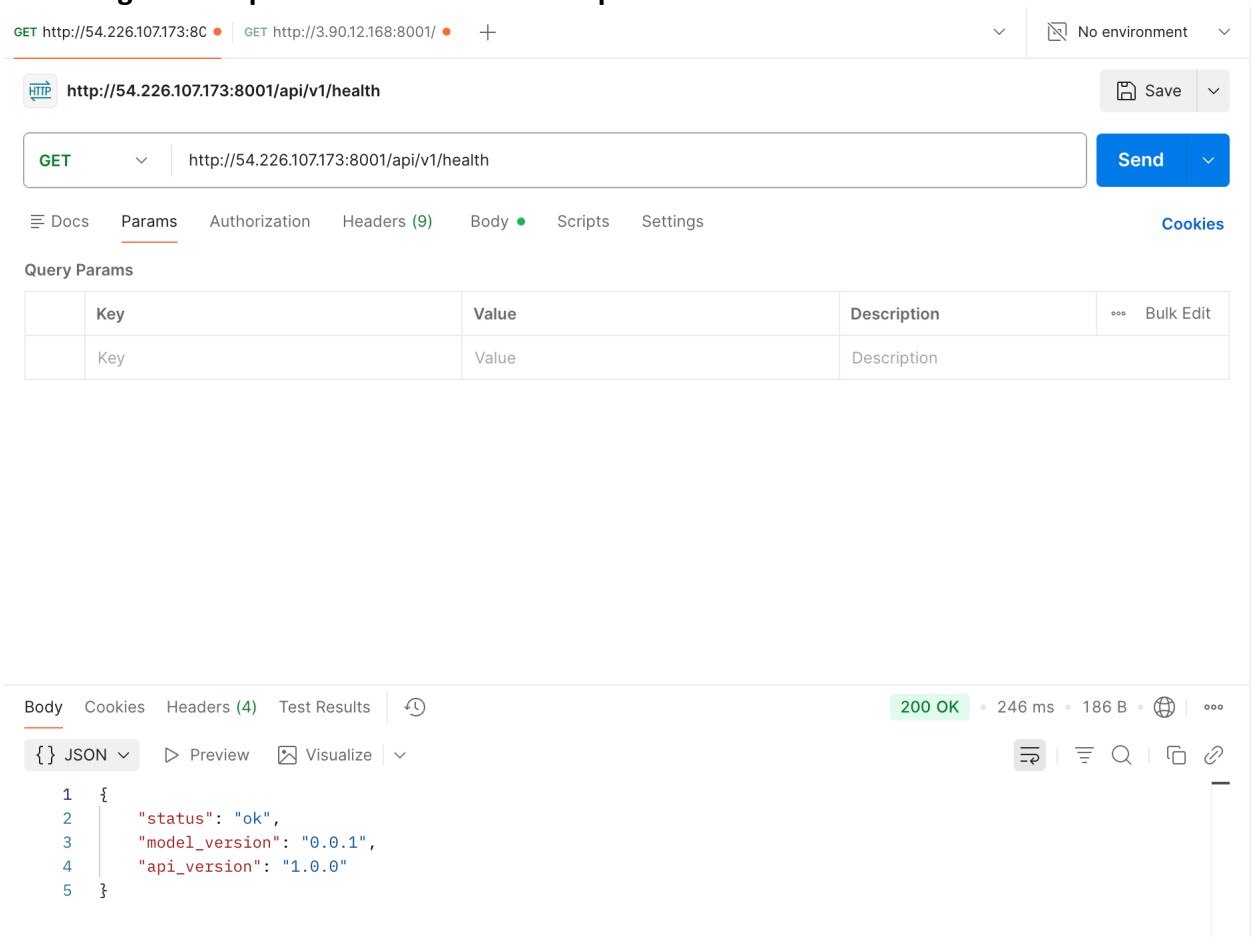
API Corriendo en el servidor:

```
api — ubuntu@ip-172-31-23-178: ~/deployment_api/deployment/api-docker — ssh -i llave/deploymp.pem ubuntu@54.226.107.173 — 146×30
subscription-api | * 'schema_extra' has been renamed to 'json_schema_extra'
subscription-api |   warnings.warn(message, UserWarning)
subscription-api | /usr/local/lib/python3.11/site-packages/sklearn/base.py:463: InconsistentVersionWarning: Trying to unpickle estimator StandardS
caler from version 1.7.2 when using version 1.8.0. This might lead to breaking code or invalid results. Use at your own risk. For more info please
 refer to:
subscription-api | https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
subscription-api |   warnings.warn(
subscription-api | /usr/local/lib/python3.11/site-packages/sklearn/base.py:463: InconsistentVersionWarning: Trying to unpickle estimator LogisticR
egression from version 1.7.2 when using version 1.8.0. This might lead to breaking code or invalid results. Use at your own risk. For more info pl
ease refer to:
subscription-api | https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
subscription-api |   warnings.warn(
subscription-api | /usr/local/lib/python3.11/site-packages/sklearn/base.py:463: InconsistentVersionWarning: Trying to unpickle estimator Pipeline
from version 1.7.2 when using version 1.8.0. This might lead to breaking code or invalid results. Use at your own risk. For more info please refer
 to:
subscription-api | https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
subscription-api |   warnings.warn(
subscription-api | INFO:     Started server process [1]
subscription-api | INFO:     Waiting for application startup.
subscription-api | 2026-03-09 23:54:40 | INFO | app.main | ==========================================================
subscription-api | 2026-03-09 23:54:40 | INFO | app.main | Iniciando Subscription Status Predictor API
subscription-api | 2026-03-09 23:54:40 | INFO | app.main | Versión API : 1.0.0
subscription-api | 2026-03-09 23:54:40 | INFO | app.main | Versión modelo: 0.0.1
subscription-api | 2026-03-09 23:54:40 | INFO | app.main | Pipeline cargado: <class 'sklearn.utils._bunch.Bunch'>
subscription-api | 2026-03-09 23:54:40 | INFO | app.main | ✓ Modelo listo para inferencia
subscription-api | 2026-03-09 23:54:40 | INFO | app.main | Documentación disponible en: http://0.0.0.0:8001/docs
subscription-api | 2026-03-09 23:54:40 | INFO | app.main | ==========================================================
subscription-api | INFO:     Application startup complete.
subscription-api | INFO:     Uvicorn running on http://0.0.0.0:8001 (Press CTRL+C to quit)
```

Desde el navegador apuntando al endpint de docs:



Peticion get al endpoint de salud a traves de postman:



Prediccion con postman

POST http://54.226.107.173:8 ● GET http://3.90.12.168:8001/ ● Runner + No environment

http://54.226.107.173:8001/api/v1/predict Save

POST http://54.226.107.173:8001/api/v1/predict Send

Docs Params Authorization Headers (10) Body ● Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify


```
{
  "inputs": [{
    "Age": 35,
    "Purchase Amount (USD)": 75.0,
    "Review Rating": 4.2,
    "Previous Purchases": 10,
    "Gender": "Male",
    "Category": "Clothing",
    "Location": "New York",
    "Size": "M",
    "Color": "Blue",
    "Season": "Summer",
    "Shipping Type": "Free Shipping",
    "Discount Applied": "Yes",
    "Payment Method": "Credit Card",
    "Frequency of Purchases": "Monthly",
    "Item Purchased": "Blouse",
    "Promo Code Used": "No",
    "Customer ID": 1
```


Body Cookies Headers (4) Test Results 200 OK 217 ms 277 B

{} JSON Preview Visualize


```
{
    "predictions": [
        {
            "customer_index": 0,
            "prediction": 1,
            "probability": 0.9474,
            "label": "Subscribed"
        }
    ],
    "model_version": "0.0.1",
    "total_records": 1,
    "errors": null
}
```